Nuclear Reliability

Forecasting Reactor Output for AI Data Centers

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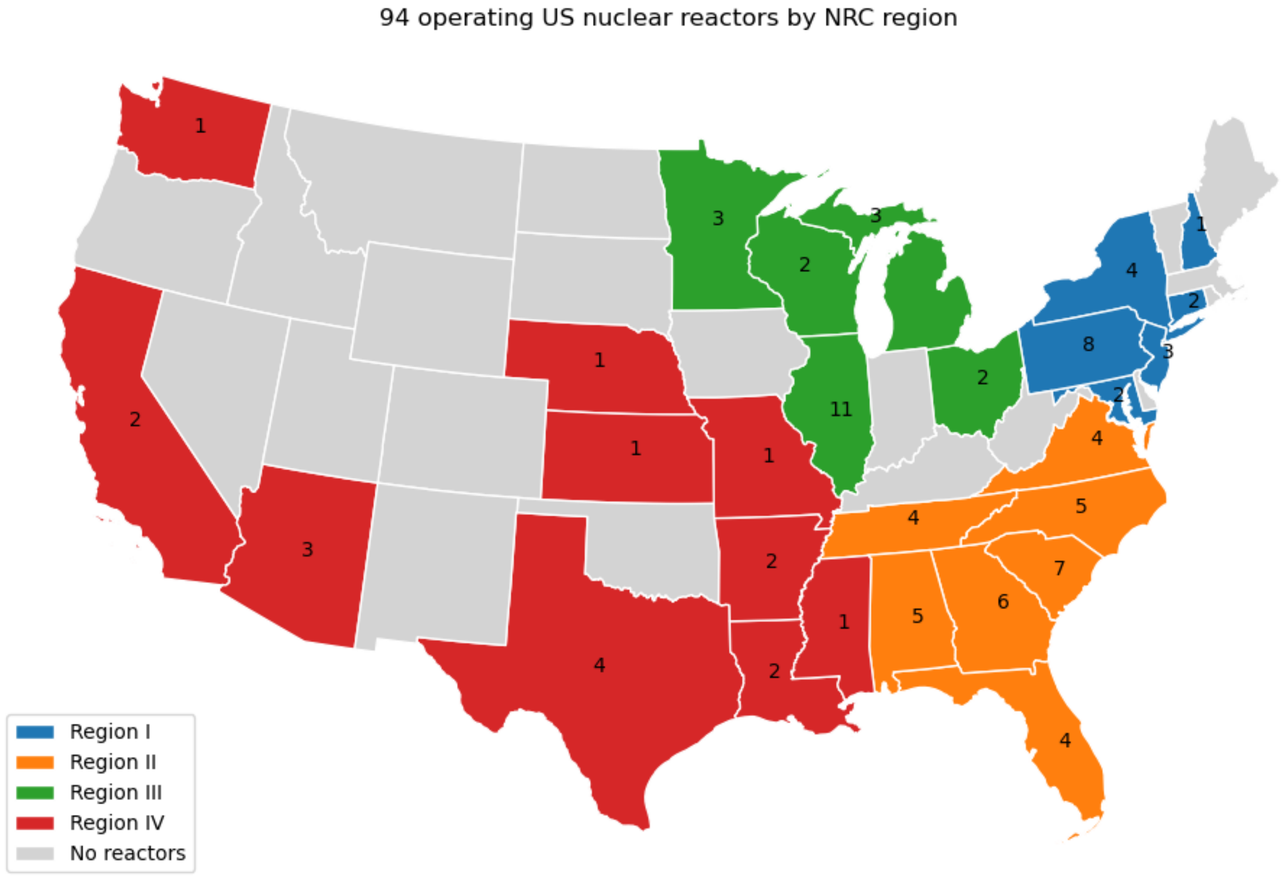
Why Reliability Matters

- Data centers need steady power around the clock
- Microsoft, Meta and Amazon have signed long-term deals with nuclear plants
- But reactors go offline for refueling, maintenance and unexpected faults
- When they do, the missing power has to come from somewhere else

Research Question

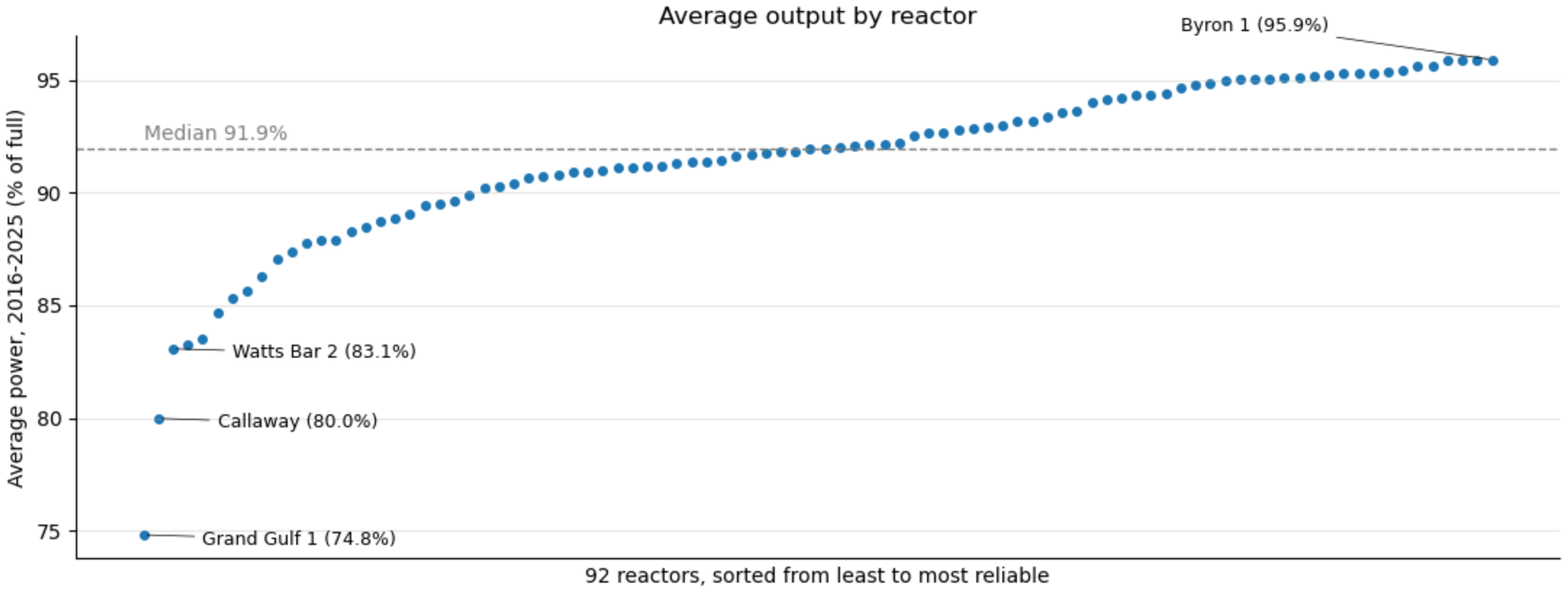
- How reliably does each US reactor deliver power?
- Can we forecast how much power it will produce over the next three months?
- Our User: a data center buyer or energy planner who depends on specific reactor but can only see public data.

Data



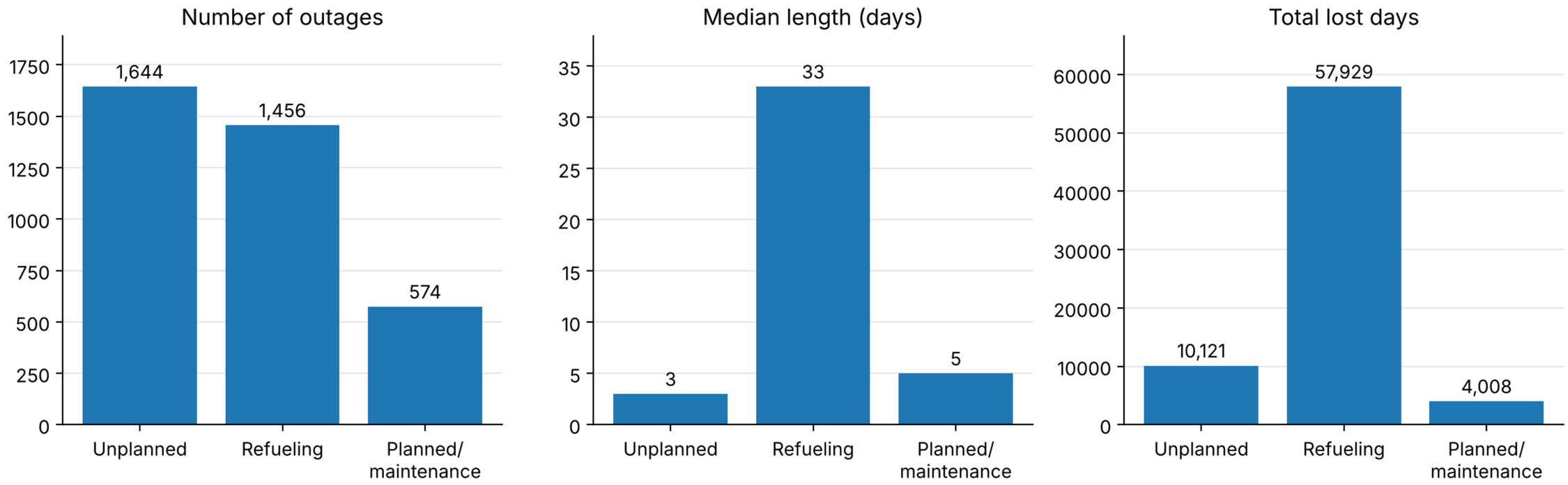
- NRC Daily Reports
- 94 operating reactors
- Daily power, 2000–2025, with reason if below full power
- Outages labeled refueling, maintenance or unplanned
- Target: average power over the next three months

How Reliable is Each Reactor?



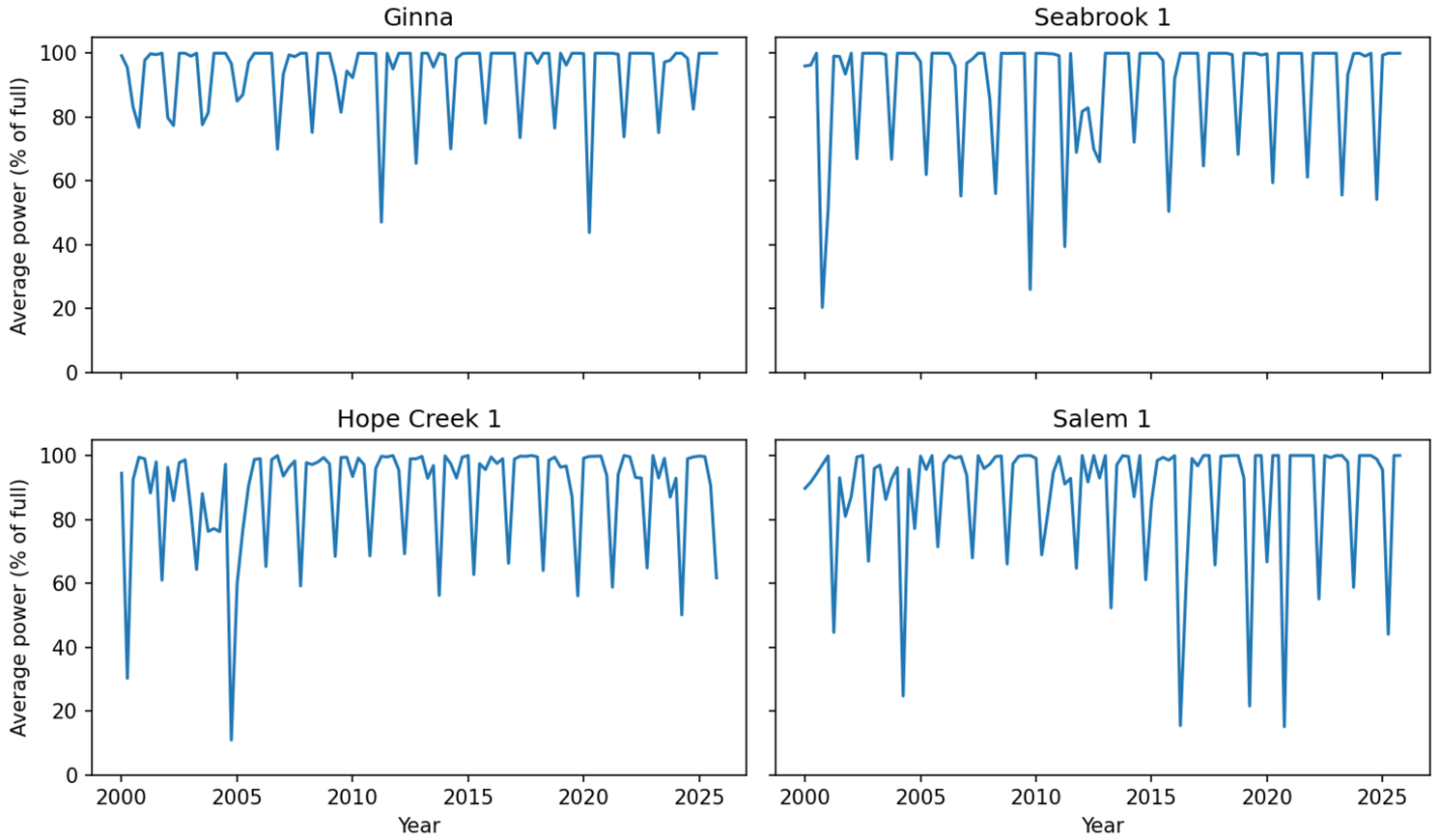
- Median reactor: 91.9% of full power
- Range: 74.8% to 95.9%, a gap of about 21 points
- On a 1,000 MW reactor that's about 210 MW of average power.

Outages by Type



Refueling is about 80% of all lost days (2000 – 2025)

Power over Time



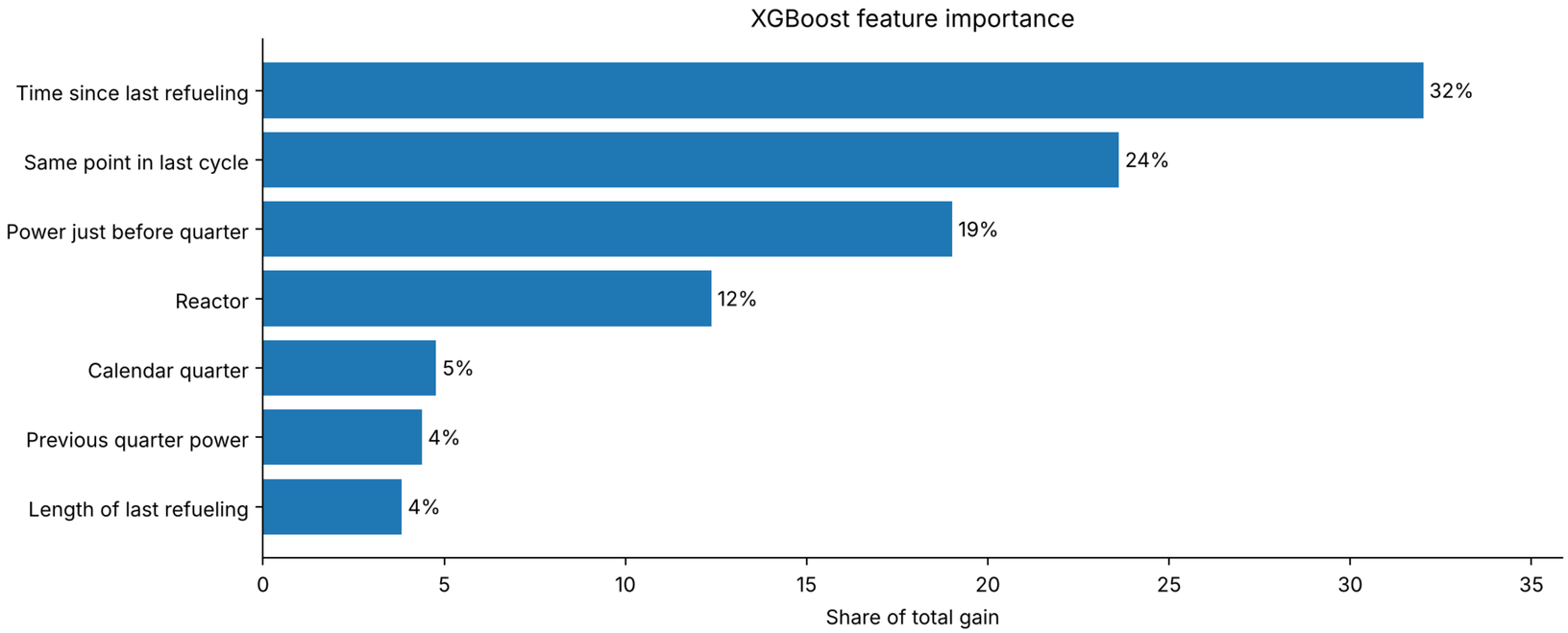
- Refueling every 18–24 months
- Dates shift
- Some outages run longer
- Surprise shutdowns

Baseline Models

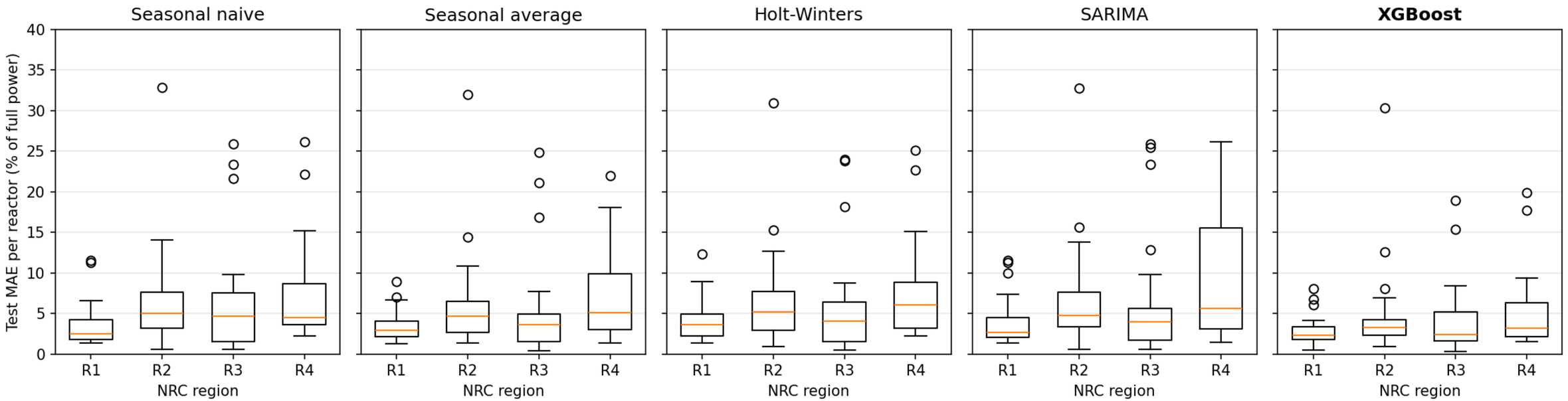
- Train: 2000–2022, Test: 2023–2025
- Seasonal naive: repeat the last refueling cycle
- Seasonal average: average the last few cycles
- Statistical Models
  - Holt–Winters (Exponential Smoothing)
  - SARIMA (Seasonal Auto Regressive Integrated Moving Averages)

XGBoost

- Baselines only see past power levels
- XGBoost is a gradient boosting method: each new tree corrects the errors so far.
- XGBoost also uses time since refueling, refueling length and recent power



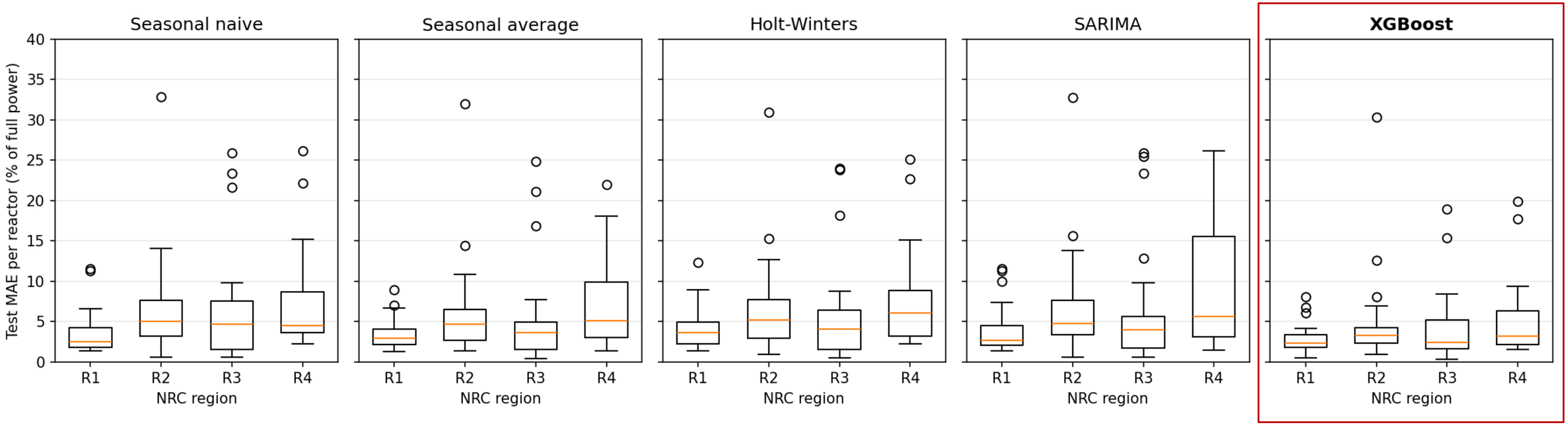
Results: Test Error by Region



- Box: spread of errors for reactors in that region
- Orange line: median. Lower is more accurate
- Circles: reactors much harder to predict

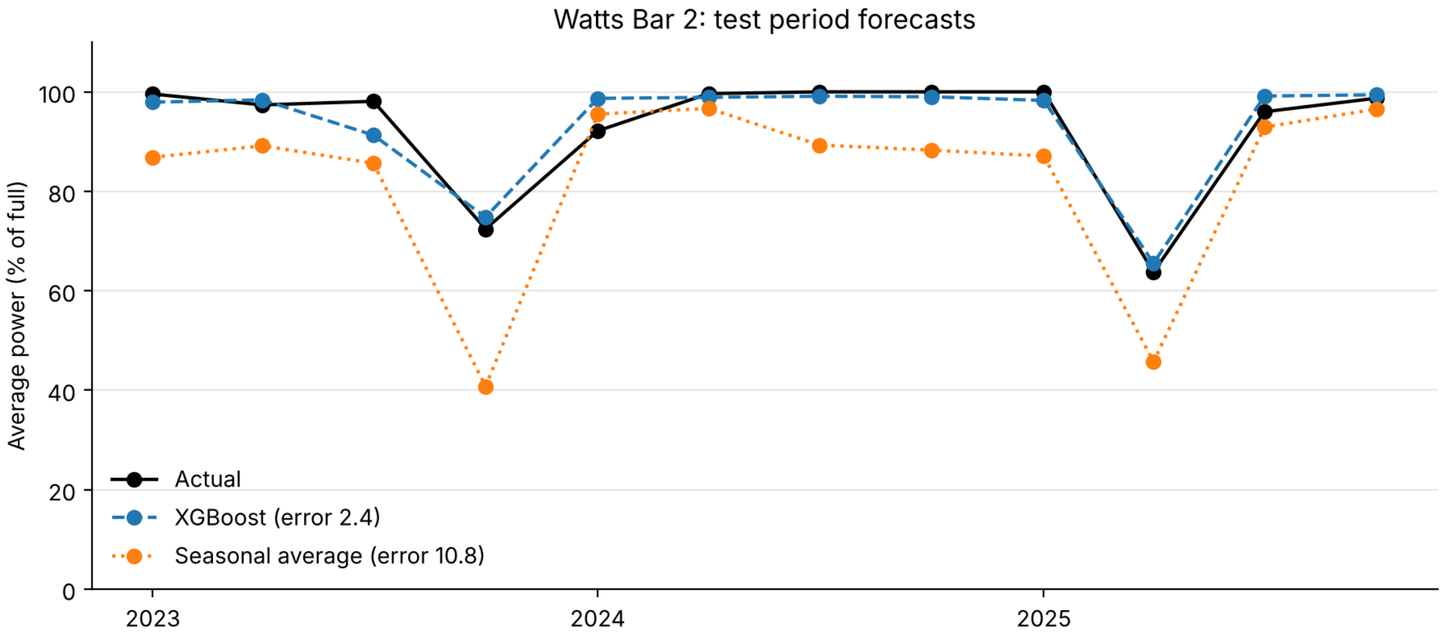
Mean absolute error per reactor, 2023–2025. 92 reactors (Vogtle 3 and 4 excluded).

Results: Test Error by Region



- XGBoost mean error: 4.31 points of full power
- Best simple method (seasonal average): 5.59
- About 23% more accurate (95% interval: 15% to 31%)

Example: Forecast vs Actual



Not typical: XGBoost beat seasonal average on 70 of 92 reactors, with a median gain of 0.65 points.

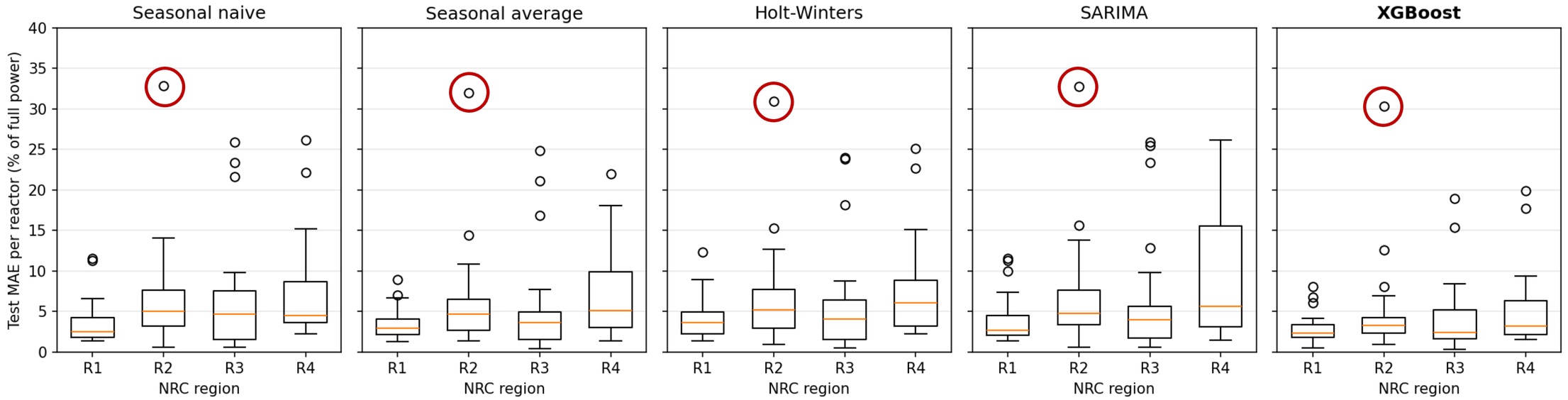
- One of the biggest improvements
- XGBoost stays close to actual power
- Seasonal average expects dips that are too deep

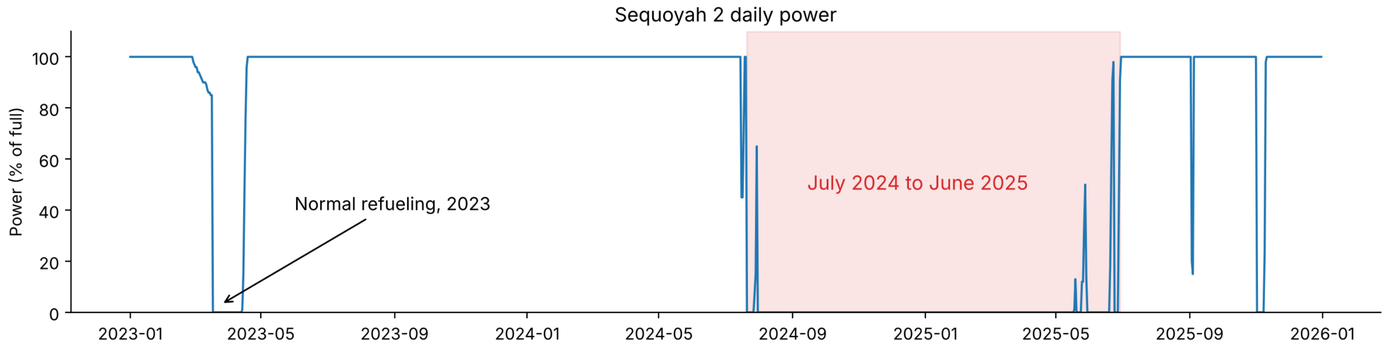
From Forecast Error to Dollars

Watts Bar 2, Q4 2023: the reactor delivered 72.42% of full power

Step	Seasonal average	XGBoost
1. Forecast	40.78%	74.81%
2. Error in points	31.65	2.38
3. Error in MW (× 1,150 MW)	363.9	27.4
4. Error in MWh (× 2,208 hours)	803,567	60,488
5. Error in dollars (× $78/MWh)	about $62.7M	about $4.7M

Hardest Reactor: Sequoyah 2





- Every model missed it (error around 30)
- Unplanned shutdown in July 2024 rolled into refueling
- Almost year-long outage

Next Steps

- Turn predicted lost power into megawatt-hours
- Multiply by electricity price forecasts for that region
- Result: cost of buying replacement power from other sources

Lost MWh × regional price = replacement cost

Thank You!

github.com/Erdos-Projects/fall-2026-nuclear-power-for-data-centers